

Alejandro Carderera

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EXPERIENCE

GitHub

Staff Applied Researcher

Remote, USA

Aug 2024 – Present

- Led applied-science strategy for GitHub Copilot Code Review; served as technical lead and acting manager, set roadmaps and priorities, aligned Product and Engineering partners, delegated work, and mentored junior researchers while sustaining team velocity through reduced capacity.
- Built and productionized an agentic comment-generation tool that delivered the product's largest quality gain from a single experiment at the time, increasing positive feedback from 70.2% to 75.6% and comments per review from 2.34 to 3.52.
- Drove a frontier-model migration through prompt and code-diff representation engineering, increasing pull requests receiving comments from 70% to 82% and comments per review from 1.6 to 2.6 without degrading quality; a redesigned diff format raised positive feedback for the weakest comment category from 42% to 74%.
- Re-architected comment filtering from a multi-stage GPT-4o and machine-learning pipeline to a single cache-aware model call, simplifying the production system while improving quality and coverage; shipped just-in-time context management that prevented token-exhaustion failures in 5.2% of agentic reviews.
- Shipped linked pull-request and issue retrieval to all users after an online A/B experiment produced a statistically significant 2.13% improvement in positive feedback with no additional token usage; drove telemetry fixes and quality gates that enabled reliable experimentation and caught regressions before release.
- Built reusable evaluation infrastructure for coding-agent memory and model-training initiatives, including approximately 50 dependency graphs and 150 tasks that unblocked two projects in under two weeks; partner-model benchmarking helped improve recall from 18% to 45% at comparable precision.

Quantfury

Quantitative Researcher in Machine Learning (Mid – Low frequency)

Remote, USA

Jan 2022 – Apr 2024

- Utilized supervised/unsupervised machine learning techniques to discover tradeable patterns in financial market data. This led to two novel trading strategies, one based on unsupervised learning, and the other based on the flow of capital of various financial instruments.
- Applied advanced machine learning methodologies to model and address technical challenges, such as changepoint detection, portfolio management, and resource allocation, transforming them into tractable optimization problems.
- Improved two of the group's existing trading strategies by refining the underlying hypothesis that underpinned why and how the strategies worked. These changes were rolled out into production and produced an average improvement in P&L of 40% while increasing maximum drawdowns by 7% on average.
- Engineered a modular Python backtesting engine for the Quantitative Research team, capable of running efficiently in an online/streaming and offline/historical fashion.
- Architected and deployed a high-performance pipeline using Azure Functions to ingest raw data and output features used in live trading, with minimal latency (~500 ms.). The pipeline handled different instrument types in parallel - such as equities, futures, commodities, and cryptocurrencies - as well as various feature configurations.

J.P. Morgan Chase

Quantitative Researcher (received full-time work offer after second internship)

NYC, USA

Summer 2020 and 2021

- Developed dynamic strategies to invest excess capital from the firm's operations to maximize shareholder value.
- Built stochastic pricing model for J.P. Morgan's deposit portfolio, outperforming benchmark with an 80x speed increase and 20% accuracy improvement.

HP Inc.

R&D Engineer (Systems Integration)

Barcelona, Spain

Aug 2016 – Jul 2018

- Solved and analyzed complex development problems at the intersection of mechanical, electrical engineering and computer science.
- Development of computer vision tools to automate product quality grading.
- Conducted experiments to enhance engineering processes, and designed machine learning tools for data analysis.

EDUCATION

Georgia Institute of Technology

Ph.D. in Machine Learning (GPA: 4.0/4.0; Advisor: Sebastian Pokutta)
Thesis: Faster Conditional Gradient Algorithms for Machine Learning

Atlanta, USA
Aug. 2018 – Dec 2021

Cornell University

M.S. in Applied and Engineering Physics (GPA: 4.01/4.3; Henry S. Sack Memorial Award)
Thesis: Drop Electrohydrodynamics

Ithaca, USA
Aug 2014 - Jul 2016

Universidad Politécnica de Madrid

B.S. in Industrial Engineering (GPA: 7.7/10)

Madrid, Spain
Aug 2010 - Jul 2014

SKILLS

LLM and Agentic Systems

Production LLM applications, agentic workflows, tool calling, context retrieval and management, prompt and system-instruction optimization, frontier-model integration, multi-model evaluation, token-cost and latency optimization

Evaluation and Experimentation

Offline evaluation, online A/B experimentation, benchmark and dataset design, model-quality assessment, statistical significance testing, production telemetry analysis, regression and quality gating

Software Engineering

Python, TypeScript (production contributions), C++, SQL, Git, Microsoft Azure Functions; production ML/LLM pipelines, reusable evaluation tooling, cross-functional technical delivery

Machine Learning and Optimization

Deep learning, supervised and unsupervised learning, reinforcement-learning model evaluation, probabilistic modeling, linear optimization, nonlinear optimization, convex analysis

BOOKS

Braun, G., Carderera, A., Combettes, C. W., Hassani, H., Karbasi, A., Mokhtari, A., and Pokutta, S. (2025). *Conditional gradient methods: From core principles to AI applications*. **Society for Industrial and Applied Mathematics**

PUBLICATIONS

Carderera, A., and Pokutta, S. (2026). *Second-order Conditional Gradient Sliding*. **Data Science and Optimization, Fields Institute Communications**

Carderera, A., Pokutta, S., Schütte, C., and Weiser, M. (2025). *An efficient first-order conditional gradient algorithm in data-driven sparse identification of nonlinear dynamics to solve sparse recovery problems under noise*. **Journal of Computational and Applied Mathematics**

Besaçon, M., Carderera, A., and Pokutta, S. (2022). *FrankWolfe.jl: a high-performance and flexible toolbox for Frank-Wolfe algorithms and Conditional Gradients*. **INFORMS Journal on Computing**

Carderera, A., Besaçon, M., and Pokutta, S. (2021). *Simple steps are all you need: Frank-Wolfe and generalized self-concordant functions*. **Proceedings of NeurIPS**

Carderera, A., Diakonikolas, J., Lin, C. Y., and Pokutta, S. (2021). *Parameter-free Locally Accelerated Conditional Gradients*. **Proceedings of ICML**

Diakonikolas, J., Carderera, A., and Pokutta, S. (2020). *Locally Accelerated Conditional Gradients*. **Proceedings of AISTATS**

AWARDS

M.H. Stewart Fellowship: Merit-based scholarship (Georgia Institute of Technology, 2018)

Henry S. Sack Memorial Award: Top academic performer (Cornell University 2014-2016)

Research Collaboration Scholarship: Merit-based scholarship (Spanish Ministry of education, 2014)

Excellence Scholarship: Merit-based scholarship (Autonomous Community of Madrid, 2020)